OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T4:
RetBot system identification, part II

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1 Preparations

1.1. (aV) Steps to solve the control task

In order to solve a control problem, a mathematical model of the controlled system is first developed and the performance requirements of the control system are defined. The controller is then designed based on this, during which the structure and parameters of the control law are specified. Finally, the controller is implemented in hardware or software and validated on the actual system. This process is illustrated in Figure 1.1 according to Lunze [2].

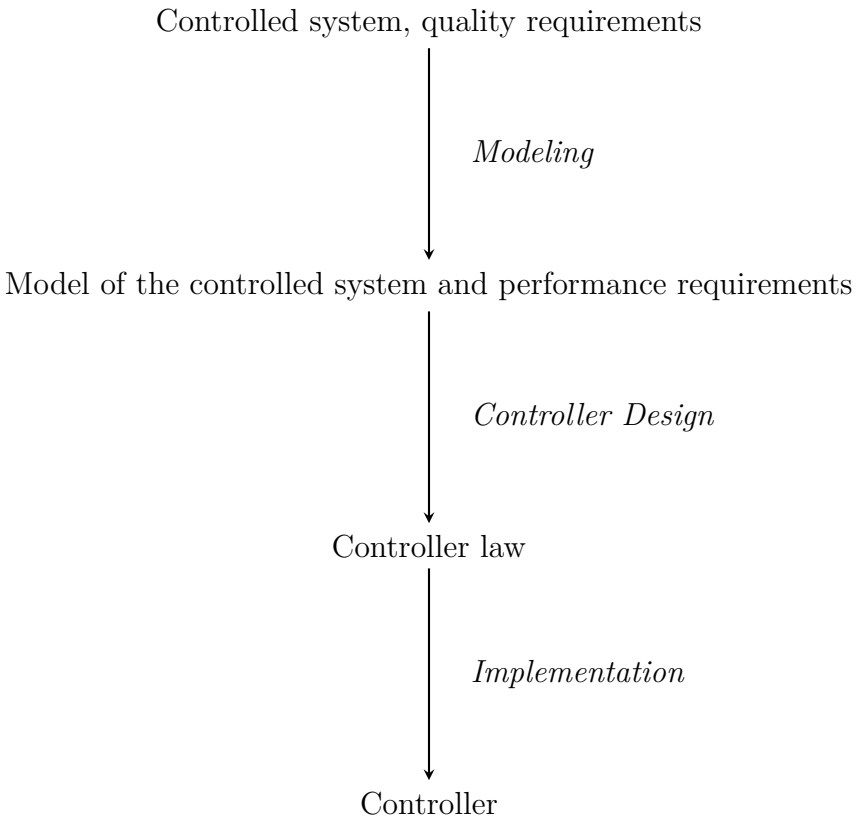


Abbildung 1.1.: Approach to control engineering problems according to Lunze [2]

1.2. (bV) Block diagram of the control loop

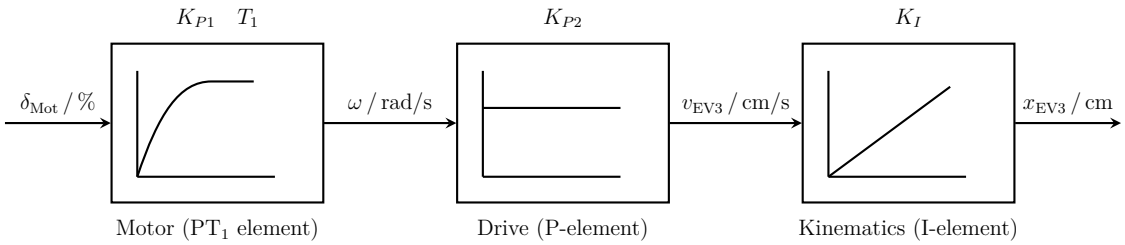
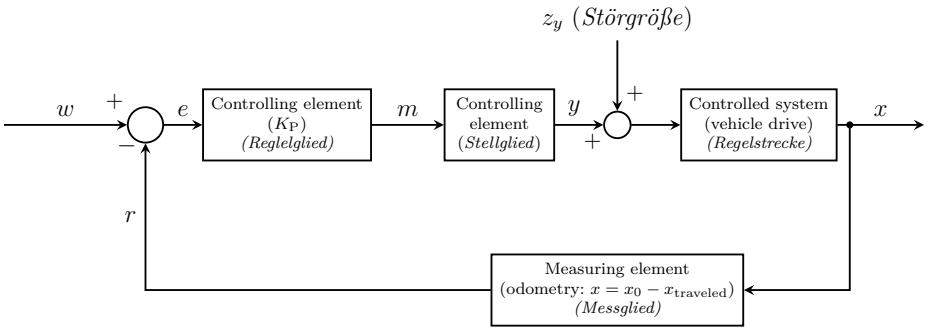


Abbildung 1.2.: The block diagram illustrates the configuration of the EV3 robot.

The control process of the EV3 robot (Figure 1.2) begins with the duty cycle δ, which is provided as an input signal in the form of pulse width modulation (PWM). This signal determines the average voltage of the DC motor and, consequently, the torque and rotational speed. The resulting rotational speed is transmitted to the wheels via the gearbox and converted into the robot's linear translational speed. Finally, the robot's kinematics describe the relationship between the rotational movements of the individual wheels and the resulting overall movement of the robot.

1.3. (cV) Block diagram of a distance control system (according to DIN IEC 60050-351).

Diagram 1.3 shows a block diagram of a distance control system using a P controller according to DIN IEC 60050-351 [1]. The system consists of a reference variable w (target distance), a control difference e, a controller output m, a manipulated variable y, a disturbance variable z_y, a controlled variable x (actual distance), and a feedback variable r. The controlling element (K_P) processes the control difference to generate the controller output, which the final controlling element converts into the manipulated variable acting on the controlled system. The disturbance z_y acts additively on the manipulated variable, and the measuring element derives the actual distance from the distance travelled, $x = x_0 - x_{\mathrm{traveled}}$, providing the feedback variable r.



w: reference variable (*Führungsgröße*, target distance)
e: control difference (*Regeldifferenz*), $e = w - r$
m: controller output (*Reglerausgangsgröße*)
y: manipulated variable (*Stellgröße*)
z_y: disturbance variable (*Störgröße*)
x: controlled variable (*Regelgröße*, actual distance)
r: feedback variable (*Rückführgröße*)

Abbildung 1.3.: Block diagram of a distance control system using a P controller according to DIN IEC 60050-351

2 Experiment T4: RetBot system identification, part 2

2.1. (d) System description

> *(d) Use the recorded data from the previous experiment and write a Matlab program that calculates the data required for the system description for both the left and right drives. Draw graphs containing the four different conditions and the tangents, ... you applied for determining the system behavior. Consider a reasonable time frame, rather than adopting a "take what's available" approach.* [3]

Each drive train is modelled as an IT_1 element with the transfer function $G(s) = K_I/[\,s\,(T_1\,s+1)\,]$, mapping the input duty cycle $\delta\,/\,\%$ to the traveled distance $x\,/\,\mathrm{cm}$ of the wheel. The two parameters — the integral gain K_I and the time constant T_1 — are extracted from the recorded step responses by an asymptotic-tangent procedure, carried out separately for the left wheel (QEI1) and the right wheel (QEI2) at each of the four duty cycles $\delta \in \{25,\,50,\,75,\,100\}\,\%$.
The QEI step counts are first converted into a traveled distance via the wheel circumference,

$$x = \frac{\text{steps}}{360^\circ} \cdot d_R \cdot \pi, \qquad d_R = 5{,}6\,\mathrm{cm}, \tag{2.1}$$

where one encoder revolution (360°) advances the wheel by its circumference $d_R\,\pi$.
For each step response $x(t)$, the asymptotic tangent $x_\infty(t) = a\,t + b$ is fitted by linear regression to the settled (linear) tail of the recording. Because the IT_1 step response asymptotically approaches the ramp $K_I\,\delta\,(t - T_1)$, the tangent slope yields the integral gain normalised per duty-cycle percentage,

$$K_I = \frac{a}{\delta}, \tag{2.2}$$

and its time-axis intersection yields the time constant,

$$T_1 = -\frac{b}{a} - t_{\text{step}}, \tag{2.3}$$

where t_{step} denotes the instant at which the duty cycle is applied. The four per-duty-cycle estimates are then averaged for each motor (Figures 2.1 and 2.2), giving the operating-point-independent parameters $\overline{K_I}$ and $\overline{T_1}$ that enter the differential equations in Section 2.2.

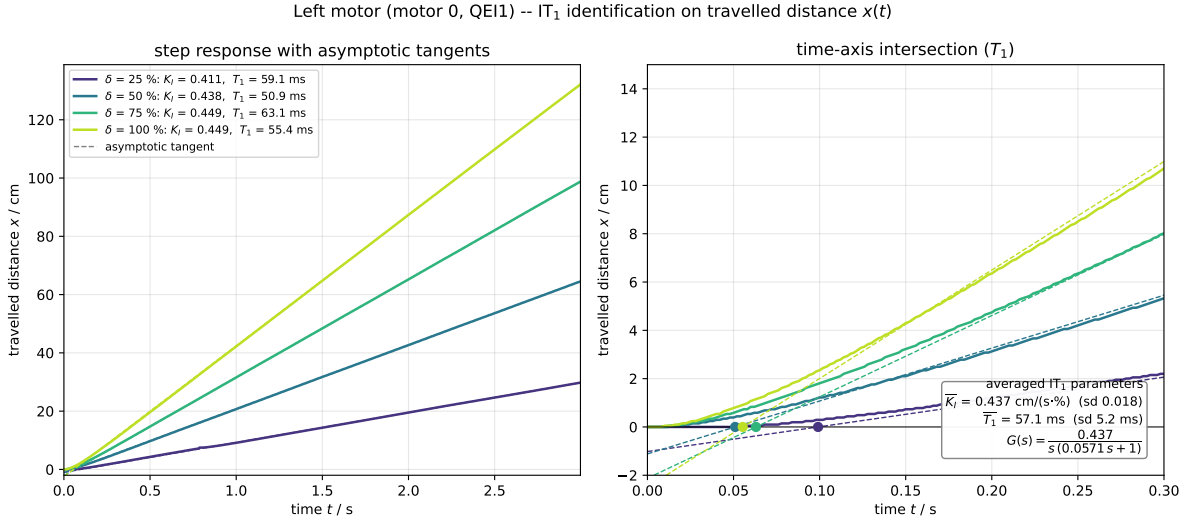


Abbildung 2.1.: Left drive (motor 0, QEI1): traveled distance $x(t)$ for all four duty cycles together with the fitted asymptotic tangents. Per-duty-cycle values of K_I and T_1 are given in the legend; the right panel zooms onto the time-axis intersections that yield T_1 and shows the averaged IT_1 parameters $\overline{K_I}$, $\overline{T_1}$, and the resulting transfer function $G(s)$.

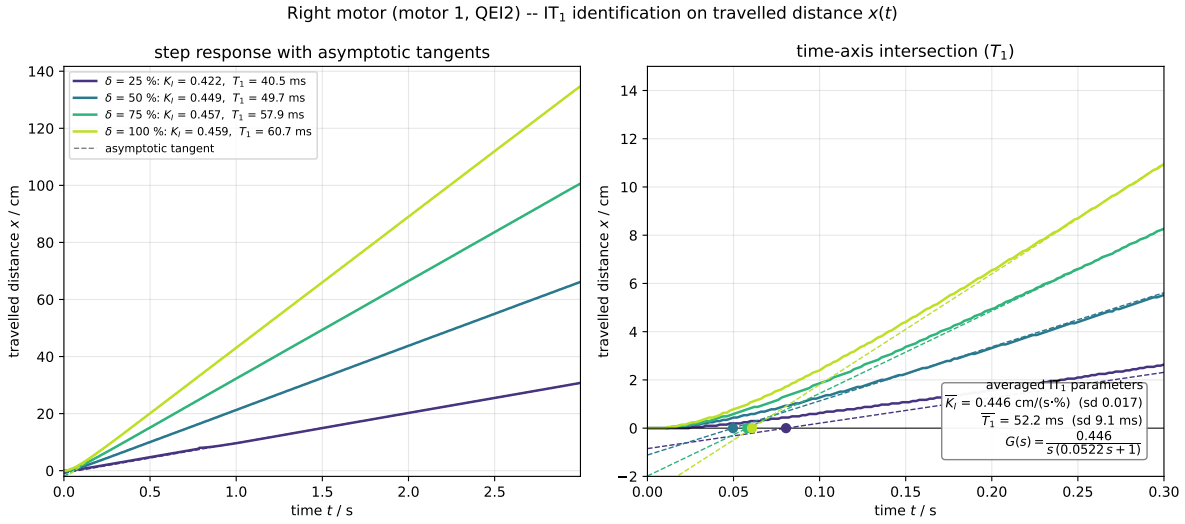


Abbildung 2.2.: Right drive (motor 1, QEI2): traveled distance $x(t)$ for all four duty cycles together with the fitted asymptotic tangents. Per-duty-cycle values of K_I and T_1 are given in the legend; the right panel zooms onto the time-axis intersections that yield T_1 and shows the averaged IT_1 parameters $\overline{K_I}$, $\overline{T_1}$, and the resulting transfer function $G(s)$.

2.1.1. Resulting Parameters

The per-duty-cycle results and their averages are summarised in Listing 2.1. The integral gain is consistent across duty cycles — the relative standard deviation stays below 5 % for both motors — which confirms the validity of the IT_1 structure within the considered operating range. The time constants scatter more strongly (up to $\sigma_{T_1} \approx 9\,\mathrm{ms}$ on the right drive), reflecting the higher sensitivity of T_1 to encoder quantisation at low duty cycles,

where the resolution of one QEI step ($\approx 0{,}049\,\mathrm{cm}$) is appreciable relative to the slow rise of the response. The same effect produces the small initial offset visible in both 25 % traces; it does not bias the identification because only the linear tail of the recording enters the tangent regression.
The two drives behave near-symmetrically: their averaged integral gains $\overline{K_{I,L}} = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}$ and $\overline{K_{I,R}} = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}$ agree within one standard deviation, and their time constants differ by less than 5 ms. The averaged parameters $\overline{K_I}$ and $\overline{T_1}$ obtained here are used directly in the differential equations and transfer functions of Section 2.2.

Listing 2.1: Identified IT_1 parameters per duty cycle and averaged values for the left and right drive.

```
motor drive   delta/%  slope_inf/cm/s  K_I/cm/(s*%)    T_1/ms
Left motor (motor 0, QEI1)  25 %   25.0000         10.2724
   0.4109   59.1133
Left motor (motor 0, QEI1)  50 %   50.0000         21.8954
   0.4379   50.9466
Left motor (motor 0, QEI1)  75 %   75.0000         33.6763
   0.4490   63.0547
Left motor (motor 0, QEI1) 100 %  100.0000         44.9191
   0.4492   55.3605
Right motor (motor 1, QEI2)  25 %   25.0000          10.5382
   0.4215   40.4807
Right motor (motor 1, QEI2)  50 %   50.0000          22.4389
   0.4488   49.6909
Right motor (motor 1, QEI2)  75 %   75.0000          34.2385
   0.4565   57.9448
Right motor (motor 1, QEI2) 100 %  100.0000          45.8991
   0.4590   60.7124

Left motor (motor 0, QEI1)
    K_I_avg = 0.4368 cm/(s*%)   (sd 0.0180)
    T_1_avg = 57.12 ms = 0.0571 s   (sd 5.18 ms)
    G(s) = 0.437 / (s (0.0571 s + 1))

Right motor (motor 1, QEI2)
    K_I_avg = 0.4465 cm/(s*%)   (sd 0.0172)
    T_1_avg = 52.21 ms = 0.0522 s   (sd 9.11 ms)
    G(s) = 0.446 / (s (0.0522 s + 1))
```

2.2. (e) Differential equations and transfer functions for both drives.

(e) Specify the descriptive differential equations and transfer functions for both the left and right drives (note: average value across the different duty cycles). Compare the step response of the identified system with the measured data to identify any inconsistencies. [3]

The following paragraphs describe the differential equations and transfer functions for both the left and right drives using constants from the previous section.
The differential equation for an IT_1 system is given by:

$$T_1\dot{x}(t) + x(t) = K_I \int x_e(t)\,dt \tag{2.4}$$

The transfer function for an IT_1 system is given by:

$$G(s) = \frac{K_I}{s(T_1 s + 1)} \tag{2.5}$$

$$h(t) = K_I \left(t - T_1(1 - e^{-\frac{t}{T_1}}) \right) \tag{2.6}$$

2.2.1. Left Motor

The parameters of the left drive (motor 0, QEI1) were identified individually for each of the four duty cycles and then averaged. The integral gain K_I was obtained from the slope of the asymptotic tangent (normalised per duty-cycle percentage), and the time constant T_1 from the intersection of that tangent with the time axis.
For the left motor, the averaged parameters are given by Figure 2.1:

$$K_{I,L} = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,L} = 57{,}1\,\mathrm{ms} = 0{,}0571\,\mathrm{s} \tag{2.7}$$

This yields the differential equation for the velocity $v_L(t)$:

$$T_{1,L}\,\dot{v}_L(t) + v_L(t) = K_{I,L}\,\delta_L(t) \quad \Rightarrow \quad 0{,}0571\,\mathrm{s}\,\dot{v}_L(t) + v_L(t) = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}\,\delta_L(t) \tag{2.8}$$

The corresponding transfer function $G_L(s)$ is:

$$G_L(s) = \frac{K_{I,L}}{s\,(T_{1,L}\,s + 1)} = \frac{0.437}{s\,(0.0571\,s + 1)} \tag{2.9}$$

2.2.2. Right Motor

The parameters of the right drive (motor 1, QEI2) were identified using the same procedure as for the left drive: K_I from the normalised asymptotic-tangent slope and T_1 from its time-axis intersection, averaged over the four duty cycles.
For the right motor, the averaged parameters are given by Figure 2.2:

$$K_{I,R} = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,R} = 52{,}2\,\mathrm{ms} = 0{,}0522\,\mathrm{s} \tag{2.10}$$

This yields the differential equation for the velocity $v_R(t)$:

$$T_{1,R}\,\dot{v}_R(t) + v_R(t) = K_{I,R}\,\delta_R(t) \quad \Rightarrow \quad 0{,}0522\,\mathrm{s}\,\dot{v}_R(t) + v_R(t) = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}\,\delta_R(t) \tag{2.11}$$

The corresponding transfer function $G_R(s)$ is:

$$G_R(s) = \frac{K_{I,R}}{s\,(T_{1,R}\,s+1)} = \frac{0.446}{s\,(0.0522\,s+1)} \tag{2.12}$$

2.3. (f) Simulink simulation of the control loop

(f) Create a Simulink simulation of the control loop designed in the preparation phase for the distance control using the identified system as the plant and a P controller. Use a gain $KP = -10\,\%\,\mathrm{cm}^{-1}$*, a target distance of* 25 cm *and an initial distance of* 80 cm *for testing. Use the integration method ode3 (Bogacki-Shampine) with a fixed step size of* 20 ms*. Do not forget the rate-limit-block (see 3.7) and use a value of* $\frac{4\,\%}{20\,ms}$*.*

Plot the reference variable, the manipulated variable and the controlled variable.

What does this sampling time mean in terms of control engineering?
Which nonlinear element do you also need to add to the simulation to simulate real conditions? [3]

The distance-control loop of Section **??** is implemented in Simulink with the two identified IT_1 plants of Section 2.2; the model is shown in Figure 2.3. The P controller uses $K_P = -10\,\%\,\mathrm{cm}^{-1}$ — negative because the measurement convention $x = x_0 - x_{\text{traveled}}$ adopted in Section **??** makes the controlled variable decrease as the robot moves towards the target. The rate-limit block bounds $|\dot{\delta}| \le 4\,\%/20\,\mathrm{ms}$, and integration is fixed-step `ode3` (Bogacki–Shampine) with $T_s = 20\,\mathrm{ms}$, reference $w = 25\,\mathrm{cm}$, and initial distance $x_0 = 80\,\mathrm{cm}$.

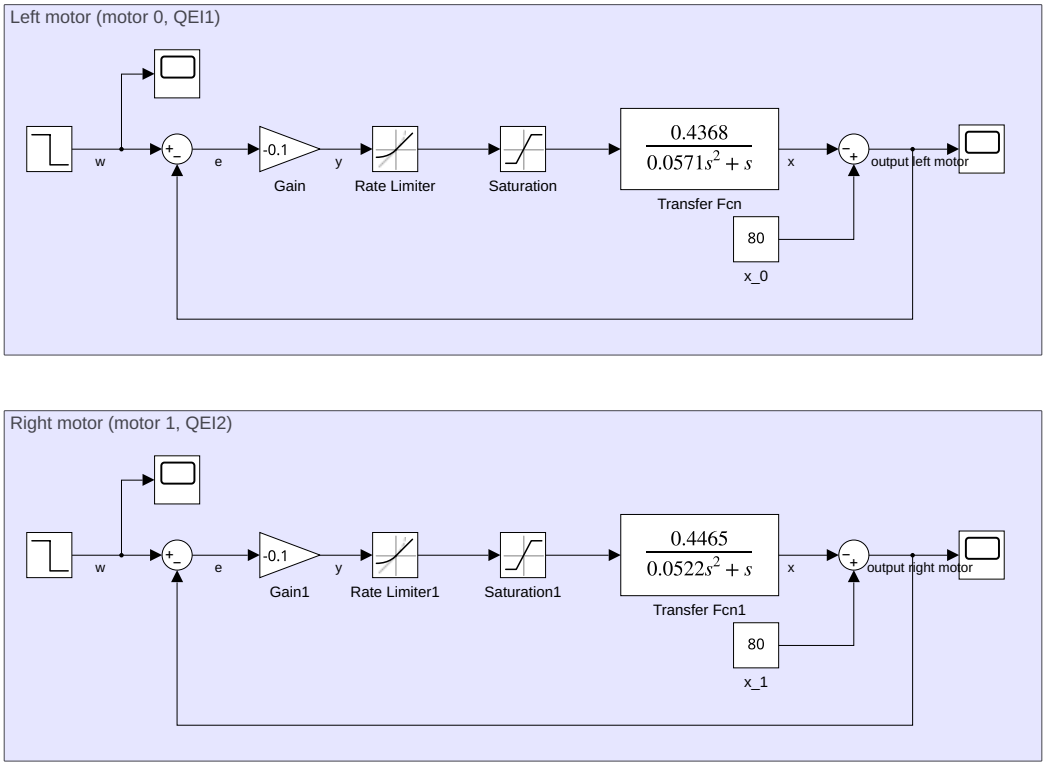


Abbildung 2.3.: Simulink model of the distance control loop with the identified IT_1 plants for both drives.

Figure 2.4 shows the simulated reference w and controlled variable x for both drives. Starting from $x_0 = 80\,\text{cm}$, x decreases monotonically toward the target $w = 25\,\text{cm}$ without overshoot, and the two drive trains track each other to within plotting accuracy. The transient is shaped by the combination of the rate-limited manipulated variable and the dominant time constant of the IT_1 plant; the steady-state error converges to zero because the plant's integrator pole supplies the type-1 behaviour the pure P controller cannot provide on its own.

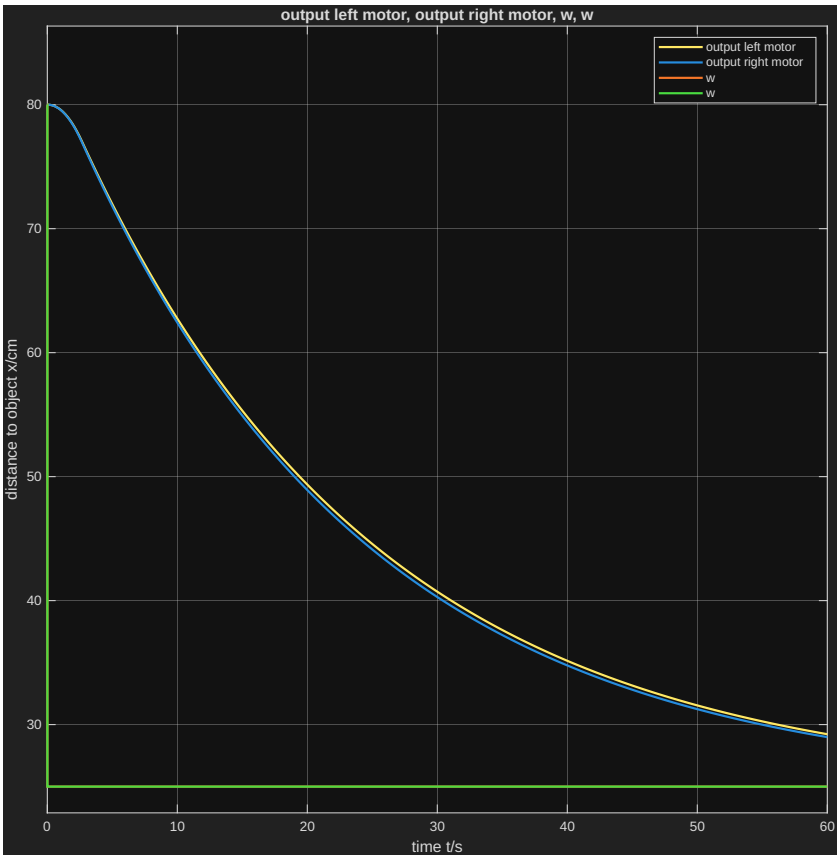


Abbildung 2.4.: Simulink output: reference w and controlled variable x for the left and right drive.

Sampling time. The fixed step $T_s = 20\,\text{ms}$ corresponds to a sampling rate $f_s = 50\,\text{Hz}$ and a Nyquist frequency of 25 Hz. Relative to the dominant plant time constant $T_1 \approx 55\,\text{ms}$ this yields $T_1/T_s \approx 3$, below the common digital-control guideline $T_1/T_s \geq 10$. The loop remains stable for the values used here, but the coarse discretisation visibly limits how finely the rate-limited manipulated variable can track the desired control law and would degrade disturbance rejection at higher frequencies.

Required non-linearity. A *saturation* block on the manipulated variable, clipping δ to the physically realisable PWM range $\pm 100\,\%$, must be added to reflect the EV3 hardware. Without it the linear P controller demands $\delta = K_P\,(w - x_0) = 550\,\%$ at

the moment of the step — a value the drive cannot produce, so the simulation would overestimate the achievable closing speed of the real system.

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

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Ort, Datum und Unterschrift

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B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time approximately 1,500 lines and a simulink model) is ***not*** included in this appendix. Including it as a verbatim listing caused major build instability again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T4-SoSe2026.git`

Literatur

[1] DIN Deutsches Institut für Normung e. V. *Internationales elektrotechnisches wörterbuch – teil 351: Leittechnik (IEC 60050-351:2013)*. Norm. Berlin, Sep. 2014. URL: https://www.dinmedia.de/de/norm/din-iec-60050-351/208013542.

[2] Jan Lunze. *Regelungstechnik 1: Systemtheoretische Grundlagen, Analyse und Entwurf linearer Regelungssysteme*. 9. Aufl. Berlin, Heidelberg: Springer Vieweg, 2016.

[3] Torsten Reitmeier. *Internship in Control Engineering (RTP)*. Praktikumsversuche. 2025.